Recall the continuity method

(M, α) compact Kähler s.t. $\alpha \in 2\pi c_1(M)$
(M is Fano)

Find ω_t solving

$$\mathrm{Ric}(\omega_t) = t\omega_t + (1-t)\alpha \qquad (*_t)$$

Suppose solution exists for $t \in [0, T)$

For $t \geq T/2$ we have $\mathrm{Ric}(\omega_t) \geq \frac{T}{2}\omega_t$

Myers' Thm $\Rightarrow$ $\mathrm{Diam}(M, \omega_t) < C_T$

$\mathrm{Vol}(M, \omega_t) = \mathrm{Vol}(M, \alpha)$ indep. of t

Volume comparison $\Rightarrow$ $\mathrm{Vol}_{\omega_t}(B(p,1)) > v$ indep. of t $\forall p \in (M, \omega_t)$

$[\omega_t] = 2\pi c_1(M)$ $\Rightarrow$ $-\sqrt{-1}\,\omega_t$ is curvature of metric h_t on K_M^{-1}

$\Rightarrow$ $\exists\, t_k \to T$ s.t.

$$(M, \omega_{t_k}) \xrightarrow{GH} (Y, d_Y)$$

and $\exists\, m > 0$ s.t. $\Phi_k : M \longrightarrow \mathbb{CP}^N$ defined using o.n. basis of K_M^{-m} w.r.t. $h_{t_k}^m$ are embeddings converging to an embedding

$$\Phi : Y \longrightarrow \mathbb{CP}^N$$

Y is a normal projective variety

$L_Y = \mathcal{O}(1)|_Y$ satisfies $L_Y|_{Y_{reg}} = K_{Y_{reg}}^{-m}$

<u>Thm</u> Assume $T=1$. Then Y admits a (singular) KE metric.

<u>Proof</u> $-\sqrt{-1}\,\alpha$ = curvature of metric H on K_M^{-1}

Eq^n $\quad \mathrm{Ric}(\omega_t) = t\omega_t + (1-t)\alpha$

can be written as

$$\omega_{t_k}^n = h_k^{t_k} H^{(1-t_k)} \quad \text{as metrics on } K_M^{-1}$$

if locally $h_k = e^{-\varphi_k}$, $H = e^{-\psi}$ $\quad \left(\begin{matrix} h_k = h_{t_k} \\ \omega_k = \omega_{t_k} \end{matrix}\right)$

then $\quad \omega_k^n = e^{-t_k\varphi_k - (1-t_k)\psi}$

As $k \to \infty$ metrics h_k^m on K_M^{-m} converge to metric h_∞^m on L_Y with locally bounded potentials.

On Y_{reg} h_∞ is metric on $K_{Y_{reg}}^{-1}$

<u>Need</u>: curvature of h_∞ is $-\sqrt{-1}\,\omega_\infty$, where

$$\mathrm{Ric}(\omega_\infty) = \omega_\infty$$

Let $p \in Y_{sing}$, $p_k \to p$, $p_k \in (M, \omega_k)$

$\exists$ holo. coords on nhoods of p_k, converging to holo. coords around p.

Using coords identify the neighborhoods (Scaling up if necessary) view each ω_k as metric on $B(0,1) \subset \mathbb{C}^n$

Coord functions z_i are holomorphic

$\Rightarrow$ by gradient est. for z_i,

$$\omega_k > C^{-1}\omega_{Euc} \quad \text{for uniform } C.$$

We also have holo. sections of K_M^{-m} near p_k with uniform bounds w.r.t. h_k^m, i.e.

$$\omega_k = \sqrt{-1}\,\partial\bar\partial\varphi_k, \quad |\varphi_k| < C$$

In the chart $\alpha = \sqrt{-1}\,\partial\bar\partial\psi_k$, and

$$\omega_k^n = e^{-t_k\varphi_k - (1-t_k)\psi_k}\,\omega_{Euc}^n$$

Need to pass to a limit as $k \to \infty$

- $\omega_k > C^{-1}\omega_{Euc} \rightsquigarrow$ upper bounds $(1-t_k)\psi_k < C$

$\Rightarrow$ psh functions

$$t_k\varphi_k + (1-t_k)\psi_k \xrightarrow{L^1_{loc}} \varphi_\infty$$

φ_∞ cannot be identically $-\infty$, since

$$\int_{B(0,1)} \omega_k^n < C.$$

ψ_k has unif. bounded Lelong numbers

$$\left(\int_{B(0,1)} \sqrt{-1}\,\partial\bar\partial \psi_k \wedge \omega_{Euc}^{n-1} < C \int_{B(0,1)} \sqrt{-1}\,\partial\bar\partial \psi_k \wedge \omega_k^{n-1} \leq C \int_M \alpha \wedge \omega_k^{n-1} \leq C' \right)$$

$\Longrightarrow$ φ_∞ has zero Lelong numbers

so $e^{-t_k\varphi_k - (1-t_k)\psi_k} \longrightarrow e^{-\varphi_\infty}$ in L^1_{loc}

- We need measures ω_k^n to converge to ω_∞^n, $\omega_\infty = \sqrt{-1}\,\partial\bar\partial\varphi_\infty$.

$$\omega_\infty^n = (\sqrt{-1}\,\partial\bar\partial\varphi_\infty)^n$$

This follows if $\varphi_k \to \varphi_\infty$ in C^α

Note that $\Delta_{\omega_k}\varphi_k = n$ $\left(\omega_k = \sqrt{-1}\,\partial\bar\partial\varphi_k\right)$

so $|\nabla\varphi_k|_{\omega_k} < C$

<u>Claim</u> we can assume that ω_k and ω_{Euc} are bi-Hölder.

(i.e. $C^{-1} d_k^\alpha \leq d_{Euc} \leq C d_k$)

Then $|\nabla \varphi_k|_{\omega_k} < C$ implies

$$\|\varphi_k\|_{C^\alpha(\omega_{Euc})} < C$$

so $\varphi_k \to \varphi_\infty$ in C^α.

i.e. $(\sqrt{-1}\,\partial\bar\partial\varphi_k)^n \longrightarrow (\sqrt{-1}\,\partial\bar\partial\varphi_\infty)^n$ as measures.

$\Longrightarrow$ $\omega_\infty^n = e^{-\varphi_\infty}$, so ω_∞ is a (singular) KE metric on Y $\square$

<u>Main consequence of this</u>: $\mathrm{Aut}(Y)$ is reductive

i.e. $\mathrm{Aut}(Y)$ is complexification of a compact Lie group.